THYMOS HEALTH

The current leading edge of precision medical care in autoimmune and inflammatory disease

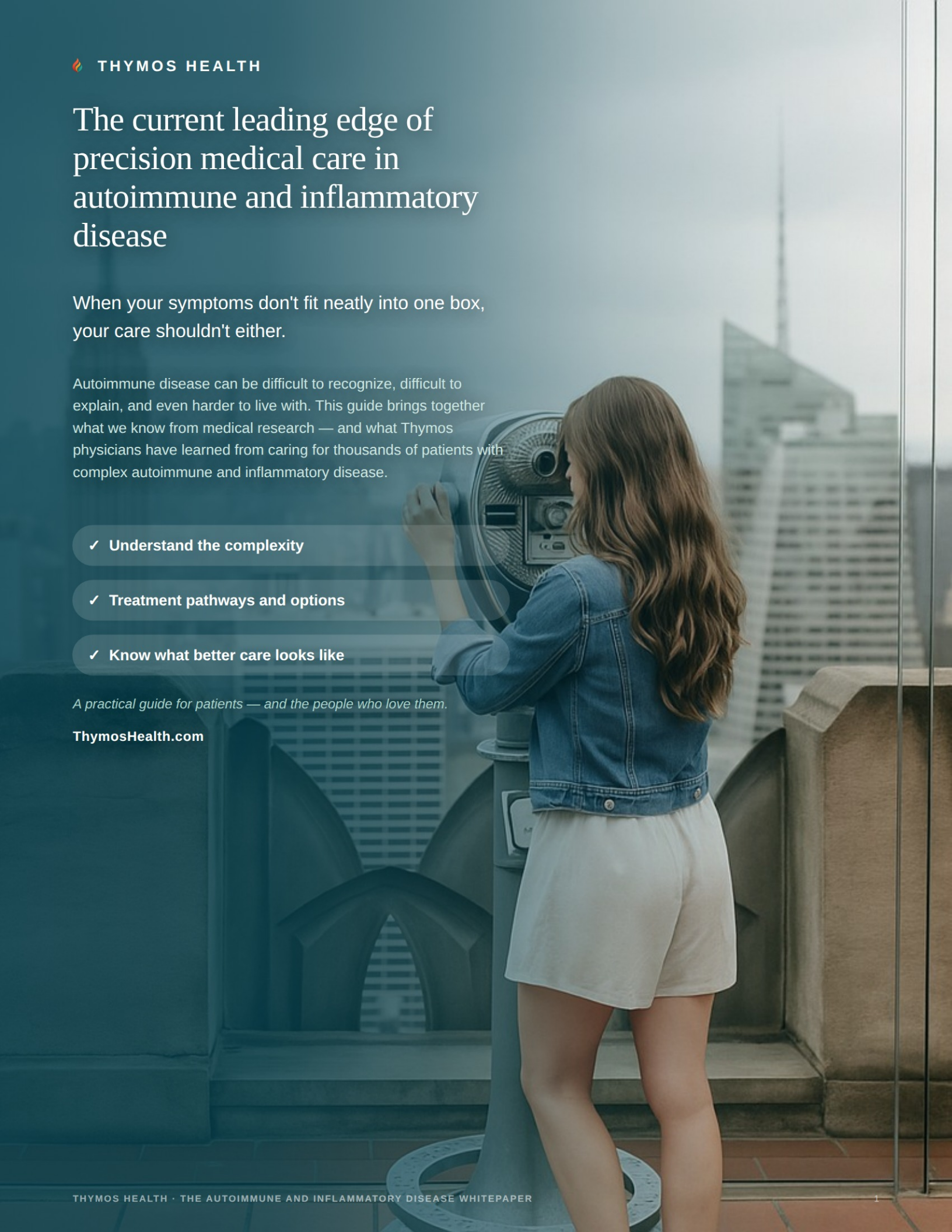
When your symptoms don't fit neatly into one box, your care shouldn't either.

Autoimmune disease can be difficult to recognize, difficult to explain, and even harder to live with. This guide brings together what we know from medical research — and what Thymos physicians have learned from caring for thousands of patients with complex autoimmune and inflammatory disease.

- ✓ **Understand the complexity**
- ✓ **Treatment pathways and options**
- ✓ **Know what better care looks like**

A practical guide for patients — and the people who love them.

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RECOGNIZING THE PATTERN

When something feels wrong, but no one can explain why — or how to address it.

Maybe you're exhausted in a way sleep doesn't fix. Your joints hurt. Your stomach is unpredictable. Your skin suddenly reacts to things it never used to. Some days you're almost yourself again — and then everything comes back.



You make appointments. You repeat your story. You get more tests. Sometimes the results are abnormal. Sometimes they're called "normal." Sometimes you're given a medication for one symptom while three others remain unexplained.

Thymos autoimmune specialists have heard this story many times — not because every autoimmune patient is the same, but because complex autoimmune disease often produces exactly this kind of experience.

> Autoimmune disease doesn't always arrive as one obvious symptom in one organ system.

SYMPTOMS THAT OFTEN SHOW UP TOGETHER

- Fatigue that doesn't go away
- Brain fog or difficulty concentrating
- Joint or muscle pain
- Digestive changes
- Rashes or skin changes
- Headaches
- Sleep disruption
- Temperature sensitivity
- Swelling or inflammation
- Numbness or tingling
- Symptoms that flare and recede
- Symptoms involving several parts of the body

Symptoms correlated with autoimmune diseases often make cross-disciplinary diagnoses and treatments challenging. That's how someone can find themselves moving between primary care, gastroenterology, rheumatology, dermatology, neurology and other specialists — with each clinician seeing an important piece of the case, but no one holding the complete picture.

If your symptoms seem disconnected, the connections and root causes may simply not have been identified yet.

UNDERSTAND THE COMPLEXITY

Autoimmune disease is common. The way it presents is anything but.

25–31M
Americans are estimated to live with autoimmune disease[1]

150
Different conditions fall within the autoimmune disease spectrum[1]

~80%
Of people diagnosed are women, though the sex distribution varies by condition[1]

34%
Of autoimmune patients in a large U.S. EHR study had more than one autoimmune disease[2]

These numbers are large. But they can still obscure what autoimmune disease feels like for an individual patient.

The immune system is a network, not an isolated organ.
When immune regulation goes wrong, its effects may appear in very different places.

The same diagnosis can look very different in different people.
Two people with the same autoimmune disease may have different symptoms, disease activity, complications and responses to treatment.

The disease can change over time.
Symptoms may flare, improve or shift. Infection, stress, sleep, hormonal changes, medications and other health factors can interact with disease activity.

More than one condition may be involved.
Autoimmune diseases can cluster — roughly one-third of people identified with autoimmune disease had been diagnosed with more than one.

Immune System

one network, many targets

Joints Pain, swelling, stiffness	**Gut** Digestion, inflammation, absorption	**Skin** Rashes, sensitivity, lesions
Brain & nervous system Brain fog, headaches, neurologic symptoms	**Heart & blood vessels** Inflammatory and cardiovascular effects	**Hormones & metabolism** Thyroid function, energy, metabolic changes

The goal isn't simply to name every symptom. It's to understand the pattern they form together.

THE DIAGNOSIS GAP

Why can it take years to get answers?

~4.5 years

The often-cited average journey to an autoimmune diagnosis[3]

~4 physicians

The number of physicians a patient may see along the way[3]

For many patients, getting diagnosed isn't one test followed by one answer. It's a process of accumulating clues.

Symptoms may appear before a disease is obvious.
Some autoimmune diseases evolve gradually. Symptoms can precede the clinical, laboratory or imaging findings needed to establish a diagnosis.

Tests need context.
A lab value isn't a diagnosis. A result inside a reference range doesn't necessarily explain persistent symptoms, just as an abnormal result doesn't automatically tell you what disease is responsible.

What does this result mean in the context of this person's symptoms, history, other tests and trajectory over time?

Different specialists appropriately look through different lenses.
A gastroenterologist evaluates the gut. A dermatologist evaluates the skin. A neurologist evaluates the nervous system. A rheumatologist evaluates rheumatic and systemic inflammatory disease.

The problem isn't specialization. The problem is what happens when nobody reconnects those findings afterward.

Symptoms evolve.
Sometimes the information needed to understand the disease doesn't exist during the first appointment. It emerges over months or years. That's why longitudinal care matters.

FRAGMENTED CARE, RECONNECTED

What you've lived through — and what your physician needs to see.

WHAT THE PATIENT EXPERIENCES

Symptoms begin
Fatigue. Pain. Digestive problems. A rash. Something feels different.

Primary care
Initial tests. Perhaps nothing conclusive.

Gastroenterology
One group of symptoms is evaluated.

Dermatology
Another symptom is treated.

Rheumatology
Inflammatory or joint symptoms are assessed.

Neurology
Brain fog, headaches or neurologic symptoms are evaluated.

More appointments. More records. More explanations. And somewhere along the way, the patient becomes the person responsible for remembering which doctor said what.

WHAT THE CLINICIAN NEEDS

One longitudinal clinical story.

Symptoms · Diagnoses · Lab trends · Imaging · Medications · Treatment responses · Specialist findings · Flares · And what changed when.

Read together — not apart.

Your medical history is data.

Autoimmune evaluation isn't only about how you feel today. Important patterns can be buried across years of medical history:

- ✓ When symptoms first appeared
- ✓ Which symptoms appeared together
- ✓ Previous infections or illnesses
- ✓ Family history
- ✓ Past diagnoses
- ✓ Prior specialist evaluations
- ✓ Medication history
- ✓ Responses to previous treatments
- ✓ Lab trends over time
- ✓ Imaging and diagnostic studies
- ✓ Nutrient deficiencies
- ✓ GI symptoms
- ✓ Sleep patterns
- ✓ Hormonal changes
- ✓ Metabolic health
- ✓ Stress and environmental exposures when clinically relevant

> A single lab result is a snapshot. Your health history is the movie.

The job of a physician managing complex disease is to understand both.

For some autoimmune diseases, uncontrolled disease activity can eventually lead to irreversible tissue or organ damage. The specific risks vary considerably by condition, which is one reason accurate diagnosis, treatment and ongoing reassessment matter.

The sooner the clinical picture becomes coherent, the sooner care can become intentional.

TREATMENT

More medicine does not automatically mean better care.

Medicine has made extraordinary progress in treating autoimmune and inflammatory disease. That innovation matters. For many patients, modern medications and biologic therapies can control disease activity, prevent complications and substantially improve quality of life.

But the scale of medication use also tells us something important about modern care.

$606B
U.S. spending on prescription medicines at estimated net manufacturer prices in 2025[4]

+10.6%
Increase in U.S. medicine spending in a single year[4]

$84B
Spent on medicines in IQVIA's immunology category in 2025 — the second-largest therapy area after oncology[5]

+12%
Growth in immunology medicine spending in 2025 alone[5]

Immunology includes many of the specialty and biologic therapies used for inflammatory diseases such as rheumatoid arthritis, psoriasis and inflammatory bowel disease.

> There has never been more treatment innovation. But prescribing the medication is only one part of treating the patient.

The right questions are also:

- **Is it the right treatment?**
- **Is it working?**
- **Is the disease still active?**
- **Are there side effects?**
- **Are other problems contributing?**
- **Does the treatment need to change?**

And who is responsible for answering those questions over time?

UNDERSTANDING YOUR OPTIONS

The treatment landscape is expanding. That makes personalization more important, not less.

Autoimmune treatment today spans a widening spectrum — from decades-old anti-inflammatories to next-generation cell therapies still being studied. Knowing roughly where a treatment sits on that spectrum, and why a physician might choose it for a specific patient, is part of understanding your own care.

SYMPTOM RELIEF

NSAIDs such as ibuprofen or naproxen ease pain and inflammation short-term but don't change the underlying disease course.

BROAD IMMUNE SUPPRESSION

Corticosteroids and conventional DMARDs calm an overactive immune system broadly — effective for many, with body-wide effects to manage.

TARGETED BIOLOGICS & SMALL MOLECULES

Biologics and JAK inhibitors aim at specific immune pathways — often highly effective, alongside infection risk and a need for close monitoring.[9]

PRECISION & CELL THERAPIES

Emerging approaches like CAR-T engineer a patient's own cells to target the disease driver directly — promising, but still early-stage and costly.[10]

Many patients move through more than one of these tiers over the course of their disease — sometimes more than once.

$87.5B
Global precision-medicine market size in 2023[10]

16.3%
Projected annual growth of precision medicine, 2024–2030[10]

$375–425K
Estimated cost of a single CAR-T infusion, illustrating why access — not just innovation — matters[10]

> More options are not the same as more clarity. Someone still has to help match the right option to the right patient at the right time.

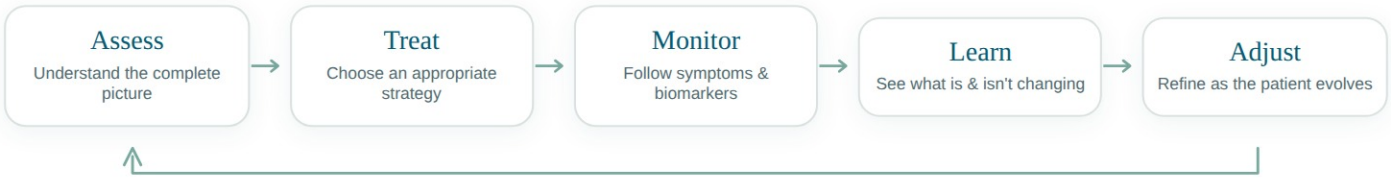
Personalized, coordinated care doesn't invent new drugs — it helps ensure a patient receives the most appropriate option for their disease activity, and helps avoid the cost, risk and burden of treatment that isn't working. An independent 2025 actuarial analysis of a personalized, physician-led autoimmune care model found meaningful reductions in avoidable utilization compared to national benchmarks: **64% fewer emergency visits, 71% less unnecessary imaging, 38% fewer overnight hospital stays and more appropriate use of high-cost specialty medications.**[11]

Precision medicine holds real promise to lower the lifetime cost of autoimmune disease — in oncology, one analysis found a single precision therapy could save health systems more than $1 billion compared to a one-size-fits-all approach.[12] But because autoimmune disease is so heterogeneous, realizing that promise today depends on something more immediate: matching the right treatment, to the right patient, at the right time.

WHAT TO EXPECT

Treatment is a process, not a prescription.

It's tempting to imagine autoimmune treatment this way: **Diagnosis → Medication → Problem solved.** Sometimes treatment is remarkably effective. But complex autoimmune care more often requires a continuing clinical loop:



A continuous loop — not a one-time decision.

The treatment plan should evolve because the patient does.

Depending on the condition and the individual, care may involve:

Managing immune activity
Prescription medications, biologic therapies or other disease-specific treatments when clinically appropriate

Looking beyond one mechanism
Evaluating other diagnoses or health factors that may contribute to persistent symptoms

Addressing relevant contributors
Nutrition, deficiencies, metabolic health, sleep, GI health when medically appropriate

Monitoring disease activity
Following symptoms, biomarkers and other clinical measures over time

Adjusting treatment
Changing the plan based on actual response — not simply the calendar

BETWEEN APPOINTMENTS

Your disease doesn't only exist during appointments.

A lot can happen between visits — a symptom worsens, sleep changes, an infection triggers a flare, or something begins working and nobody knows unless the patient remembers to mention it months later. These changes aren't interruptions to the care plan. They are clinical information.

Worsening symptoms · **New symptoms** · **Medication side effects** · **Infections** · **Sleep changes** · **GI changes** · **Fatigue** · **Flare patterns** · **Changes in function** · **Changes in disease activity**

The best autoimmune care doesn't ask only *"How are you today?"* — it tries to understand *"What has happened since we last talked — and what does it tell us?"*

> The best treatment plan isn't static. It evolves with you.

ACCESS MATTERS, TOO

Having a treatment doesn't always mean being able to receive it.

The United States is spending more on medicine than ever before. Yet patients still face significant barriers to accessing treatment.

$110B

Patient out-of-pocket spending on medicines in 2025 — a record high[4]

Nearly 2 in 3

Prescriptions for newly launched medicines go unfilled during their first year, per IQVIA, amid coverage and affordability barriers[4]

Meanwhile, PwC projects commercial medical costs will rise another 9% in 2027 — the highest medical cost trend in 17 years.[6] Pharmacy spending is one of the forces pushing costs higher.

For patients and families, those trends can show up as much more personal questions:

- **Will my insurance cover this?**
- **Can I afford the medication?**
- **Do I need another authorization?**
- **What if the treatment isn't working?**
- **Is there another appropriate option?**
- **Who helps me navigate this?**

Cost should never determine what a physician believes is medically right for an individual patient. But affordability, coverage and access are real parts of the patient's experience — and good care has to acknowledge them.

A GUT CHECK

How do you know whether your autoimmune care is working?

Ask yourself the questions on the left. Then take the matching questions on the right to your physician — they aren't about challenging your care, they're about making the thinking behind it clearer. 📋 Save or screenshot this page

ASK YOURSELF	ASK YOUR PROVIDER
Do I understand the working diagnosis? Do I know what my physician believes is causing or contributing to my symptoms?	**1.** What do you believe is causing or contributing to my symptoms? **2.** Which findings support that diagnosis? **3.** What else could explain the symptoms that remain unexplained?
Is anyone looking at the whole case — and who owns it? Are my diagnoses, medications, symptoms and specialist findings being interpreted together? If five physicians are involved, who understands what all five are doing?	**4.** Are there findings in my history that become more meaningful when looked at together? **5.** Who is coordinating the recommendations from my different specialists?
Am I actually getting better? Not simply: Am I taking treatment? But: Are my symptoms, function and disease activity meaningfully improving?	**6.** How will we know whether this treatment is working — and what should improve first?
Is my progress being measured? Are the symptoms, biomarkers or other measures that matter for my condition being followed over time?	**7.** Do changes in my labs over time tell us more than any single result? **8.** When should my disease activity and treatment response be reassessed?
What happens when something isn't working? Does someone reassess the diagnosis and treatment strategy — and can the plan actually adapt?	**9.** What symptoms or side effects should make me contact you or reconsider the plan? **10.** If I'm not meaningfully better, what do we do next?

> Being on treatment and being well managed are not necessarily the same thing.

If you're still struggling, that doesn't automatically mean there are no other options — it may mean the case deserves another look. Write the answers down; they become part of your medical history, too.

THE IDEAL MODEL

What better autoimmune care can look like.

One physician understands the whole case.
Symptoms, history, labs, medications, previous evaluations and treatment responses are interpreted together.

Your history is reconstructed.
Years of fragmented information become one longitudinal clinical story.

The diagnosis is continually tested against the evidence.
As new information appears, the clinical picture becomes clearer.

Treatment is personalized.
Clinical decisions reflect the diagnosis, disease activity, symptoms, treatment response and individual needs.

Progress is measured.
The question isn't simply whether treatment was prescribed. It's whether the patient is getting better.

The plan is reassessed.
If something isn't working, the answer shouldn't automatically be to wait another six months.

Specialists contribute to a shared picture.
The patient shouldn't have to be the only person connecting everybody else's recommendations.



You deserve care that looks at the entire picture — not just the next symptom.

THE THYMOS APPROACH

We treat the case, not just the diagnosis.

Thymos Health was built for patients whose health cannot be understood in a 15-minute snapshot.

Our physicians bring together symptoms, medical history, diagnostics, medications, previous specialist care, treatment response and ongoing disease activity into one coordinated view.

But there is another kind of experience that matters here: **Experience listening to patients.**



We've cared for thousands of people with complicated, multisystem symptoms.

> Patients who arrive with folders of records.
>
> Patients who can recite years of lab values from memory because they've had to become their own care coordinator.
>
> Patients who stopped mentioning certain symptoms because nobody seemed to think they were connected.
>
> Patients who finally received a diagnosis but still didn't feel well.
>
> Patients whose family members could see how much they were struggling even when the outside world couldn't.

We know how exhausting it is to keep explaining that something still isn't right. That experience shapes how we practice medicine.

THE THYMOS APPROACH, CONTINUED

How we put it into practice.

1 **Reconstruct**
Organize years of symptoms, diagnoses, tests, medications and specialist visits into one clinical timeline.

2 **Contextualize**
Interpret each finding against the patient's full history, disease activity and other clinically relevant factors.

3 **Personalize**
Develop a treatment strategy around the actual patient — not simply a diagnostic label.

4 **Reassess**
Track symptoms and clinical data so physicians can see whether treatment is meaningfully changing the trajectory.

5 **Coordinate**
Help ensure the patient isn't left managing a collection of disconnected specialists alone.



- One whole-system autoimmune specialist

Medical history · Diagnostics · Symptoms · Disease activity · Medications · Nutrition · GI health · Sleep · Metabolic health · Stress & environment

All interpreted as parts of the same patient.

80%+
of patients experienced meaningful symptom improvement within three months, in Thymos' internal real-world outcomes analysis[7]*

9/10
patient satisfaction score reported in Thymos internal program data[7]*

*Internal real-world outcomes. Individual responses and time to improvement vary. These findings are not a guarantee of results.

Different patients. Different diseases. One principle: understand the whole case.

These two de-identified Thymos patients started in very different places — one searching for a diagnosis, one already treated but still symptomatic. Both needed their case understood as a whole.

FIVE SPECIALISTS. YEARS OF SYMPTOMS. NO DURABLE IMPROVEMENT.

A 57-year-old patient had lived for years with worsening fatigue, brain fog, recurrent hives, joint and muscle pain and IBS-like symptoms — evaluated across primary care, gastroenterology, rheumatology, orthopedics and neurology without lasting improvement.

The symptoms weren't new. Seeing them together was.

Reconstructing her history as one clinical story surfaced ME/CFS, iron-deficiency anemia, vitamin D deficiency, subclinical hypothyroidism, dyslipidemia and a disorder of protein metabolism — then a coordinated plan, reassessed over time.

"I lost 8 pounds and was able to stop my pain medications."

Outcome: ~50% symptom reduction within 3 months; most symptoms resolved within a year, without a biologic.

What changed: One physician connected the findings across systems and kept reassessing.

DIAGNOSED. MEDICATED. STILL SICK.

A 47-year-old patient on methotrexate and hydroxychloroquine still had joint pain, rashes, cold extremities, fatigue, mouth ulcers, dry eyes and low-grade fevers — seen by primary care, two dermatologists and a rheumatologist, but not getting well.

The goal wasn't a new diagnosis. It was understanding what was still being missed.

Reassessing the whole case revealed undifferentiated connective tissue disease alongside nutritional anemia, vitamin D deficiency, elevated A1c, hyperhomocysteinemia and hyperferritinemia — then followed longitudinally as the plan evolved.

"I was finally able to do the physical activities I used to enjoy."

Outcome: ~65% symptom reduction within 6 months; tapered off immunosuppressives within a year under her rheumatologist's supervision.

What changed: Treatment became an actively managed process, not a static prescription.

Two very different cases. The same lesson.

One patient searched for an explanation. The other had a diagnosis and medication but remained symptomatic. For both, the next step was making the case coherent:

1
Reconstruct
Bring fragmented symptoms and records into one timeline.

2
Contextualize
Interpret findings against the whole patient picture.

3
Coordinate
Address relevant pieces together, not in isolation.

4
Reassess
Refine care as the patient's condition changes.

> Complex autoimmune disease deserves more than fragmented care.

These are de-identified patient examples.[8] Individual results vary and are not a guarantee of outcomes for other patients. Medication changes should be made by the appropriate treating clinician based on individual disease activity and clinical judgment.

A CLEARER PATH FORWARD

You don't have to keep starting over.

Living with autoimmune disease can change things other people rarely see.

> How much energy you have when you wake up.
>
> Whether you can concentrate at work.
>
> Whether you can make plans without wondering how you'll feel that day.
>
> Whether eating will make you sick.
>
> Whether you have enough energy for your children.
>
> Whether your partner understands why you're cancelling again.
>
> Whether you trust your own body.

And then there is the work of being a patient itself: appointments, records, insurance, labs, prescriptions, specialists, follow-ups — repeating the same story again.

Your care shouldn't become another illness you have to manage.

Complex disease requires someone willing to understand the complete case — and stay with it as the case evolves. That's what Thymos was built to do.

Ready for a clearer path forward?

Work with a Thymos autoimmune specialist.

THYMOSHEALTH.COM

FOR SOMEONE YOU LOVE

Did someone come to mind while you were reading this?

Maybe it's your daughter who is always exhausted. Your partner who has been seeing specialist after specialist. Your mother who says she's fine but no longer does the things she used to love. Your friend who finally received a diagnosis but still isn't getting better. Sometimes the person struggling with chronic illness no longer has the energy to search for another answer. You can help by sending them one.

Share this guide with someone whose symptoms still don't make sense, whose treatment isn't giving them the relief they hoped for, or who simply needs to know that a different approach to autoimmune care exists.

SHARE THIS GUIDE →

Free to read. No account required.

SELECTED SOURCES & DISCLAIMER

Where these facts and figures come from.

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12. Sano Genetics.
The Future of Precision Medicine in Autoimmune and Inflammatory Disease: A Strategic Framework. Citing a precision-oncology cost-modeling study estimating savings from precision therapy versus a one-size-fits-all treatment approach.

Important information. This guide is intended for general education and to help patients and families become better prepared to participate in their healthcare. It does not provide medical advice, diagnosis or treatment and is not a substitute for evaluation by a qualified healthcare professional.

Autoimmune diseases vary substantially in presentation, severity, prognosis and treatment. Statistics describing autoimmune diseases as a group may not apply to any particular condition or individual.

Individual treatment outcomes vary and are not guaranteed. Thymos patient outcomes described in this guide reflect internal real-world data and individual de-identified patient experiences. If you are experiencing a medical emergency, call 911 or seek immediate medical care.